
Study of the surface texture evaluation with 2 dimensional discrete Fourier transform

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(1998)

2 dimensional discrete Fourier transform (2D DFT)

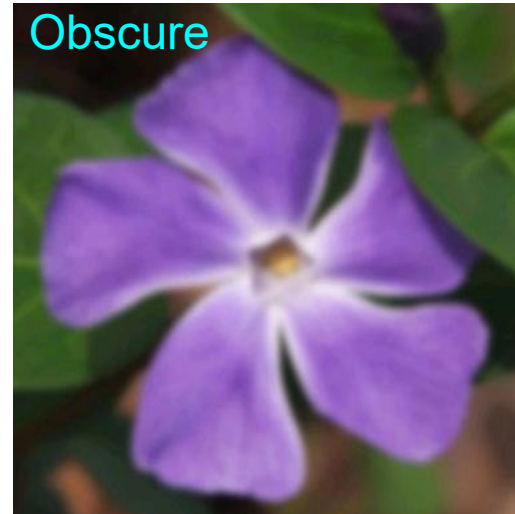
(1) Introduction 2D DFT is a very popular method in the image processing, such as filter effect. By using 2D-DFT, we can analyze 2 dimensional signals, such as roughness, surface texture, and morphology of thin film (AFM data).

Digital filter processing



Example : original

Decrease or elimination
of high frequency of 2D-DFT
spectrum

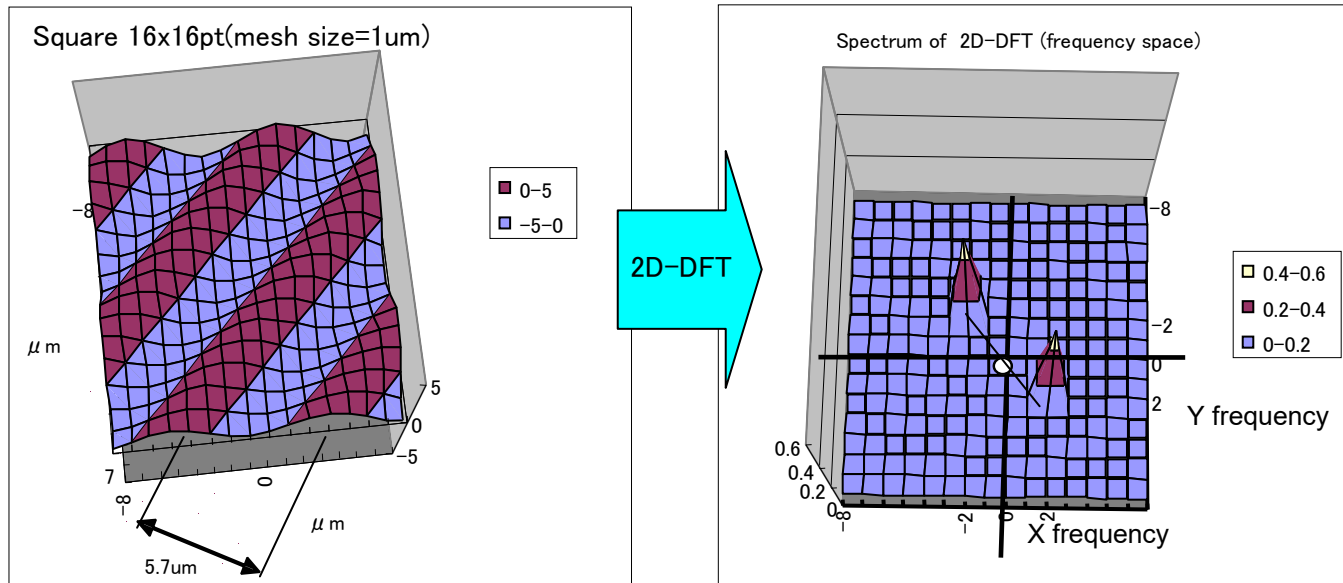


Enhancement in intensity
of high frequency of 2D-DFT
spectrum

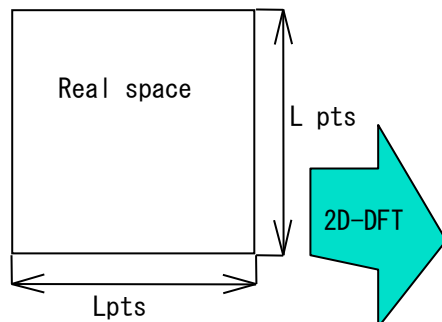


2 dimensional discrete Fourier transform (2D DFT)

Sample1 : slant(45deg.) direction, amplitude 1 μ m, wave length 5.7 μ m, only 1 sin curve [made by calculation]



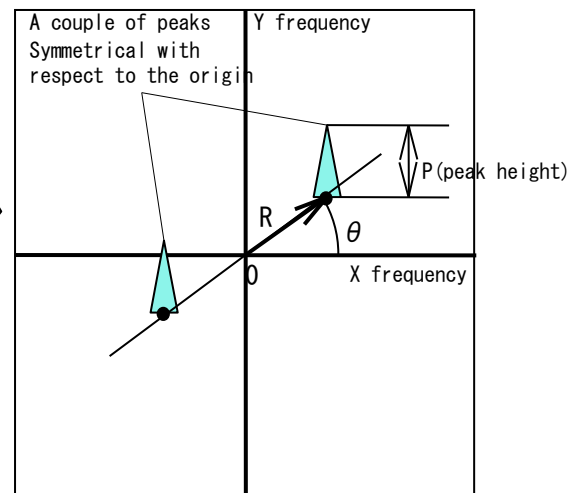
DATA size of measurement



P : peak height
 R : Radius (Frequency of wave)
 θ : peak angle

Depth of sin curve = $2 \times P$
 Direction of wave = θ
 Wave length = $(L/R) \times \text{mesh size}$

2D DFT out put (spectrum)



We can analyze by using spectrum

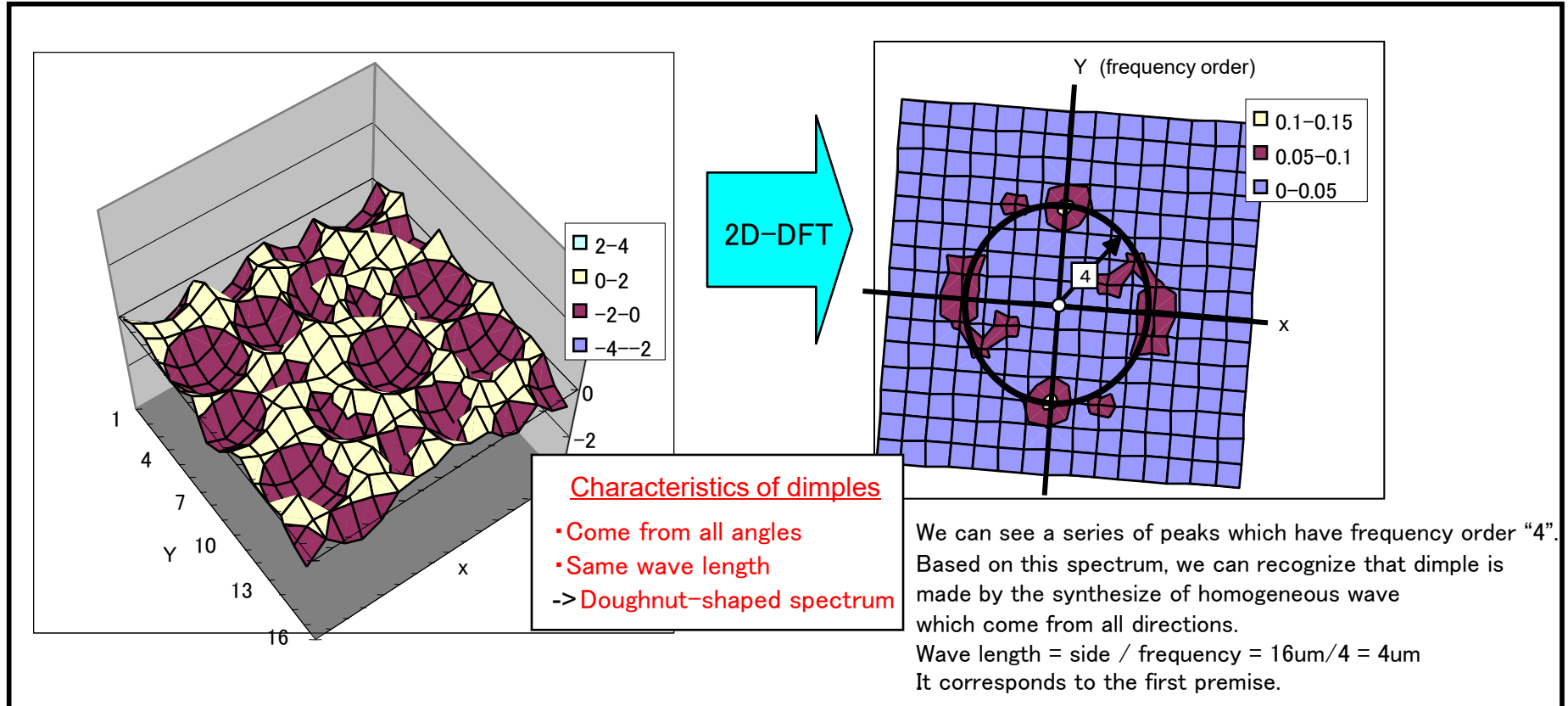
- 1) Peak height $\rightarrow$ Depth of the wave
- 2) Peak angle $\rightarrow$ Direction where wave come from
- 3) Radius from origin $\rightarrow$ Wave length

For example....sample1

Wave direction : upper left 45deg
 Amplitude: $P \times 2 = 0.5 \times 2 = 1\mu$ m
 Wave length:
 frequency (R) = $(2^2 + 2^2)^{0.5} = 2.83$
 Length = $L/R = 16/2.83 = 5.7\mu$ m

2 dimensional discrete Fourier transform (2D DFT)

Sample2 : dimple Dia.=4 μ m,made by using part of sine curve, placed at random
Artificial surface model, rectangle 16 x 16 μ m (mesh size = 1 μ m)



Note : This example is some ideal case. A slant of the surface and waviness are eliminated.

In many cases, we will find a series of noise on the X and Y axes, caused by those factors.

It is one of the major issue when we treat some practical case. We must take care of treatment of noises.

2 dimensional discrete Fourier transform (2D DFT)

Surface texture : Study of dimple evaluation method.

Measurement

It is very difficult to evaluate periodicity of surface texture by using conventional 1 dimensional roughness measurement. We have tried investigation on 2 dimensional surface measurement, by using recent roughness instrument and application of digital signal analysis. Therefore, we tried to find characteristic and difference of each texture.

Elimination of slant and waviness

2D DFT

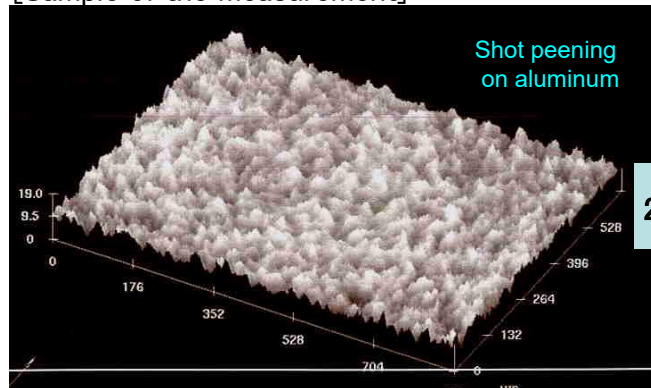
Statistics of spectrum

Accumulated by wave length
Accumulated by angle

Identify peak position

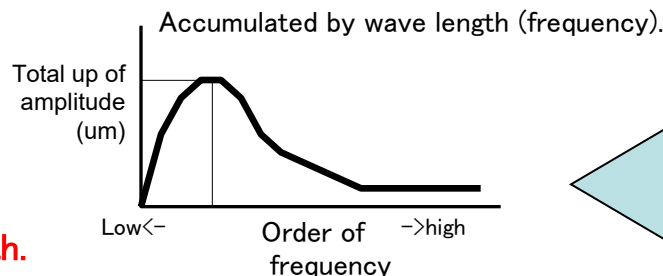
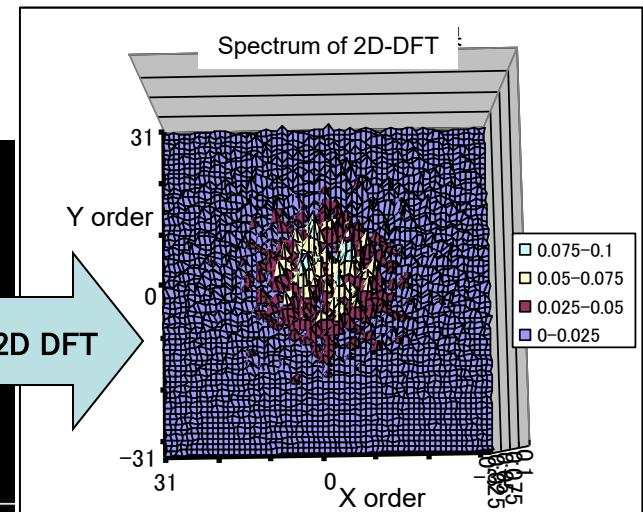
Peak pos. relates to dimple size.
Peak height relates to dimple depth.

[Sample of the measurement]



Shot peening
on aluminum

2D DFT



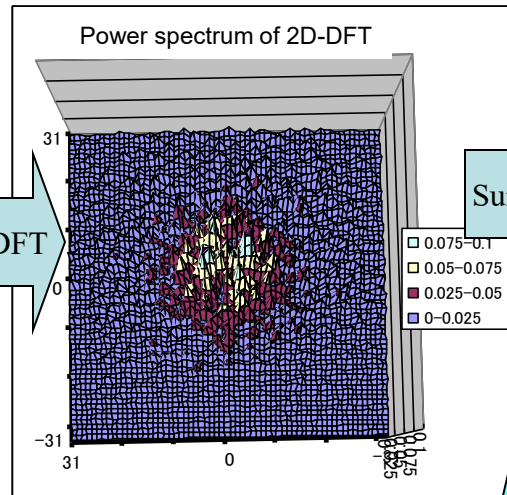
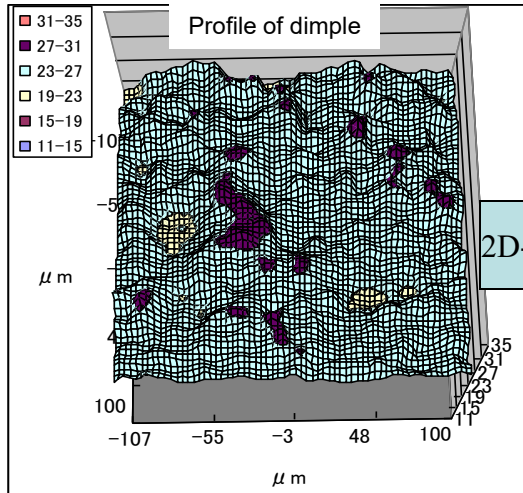
2D-DFT Signal and noise

Sample spectrum (Shot-peening on Al)

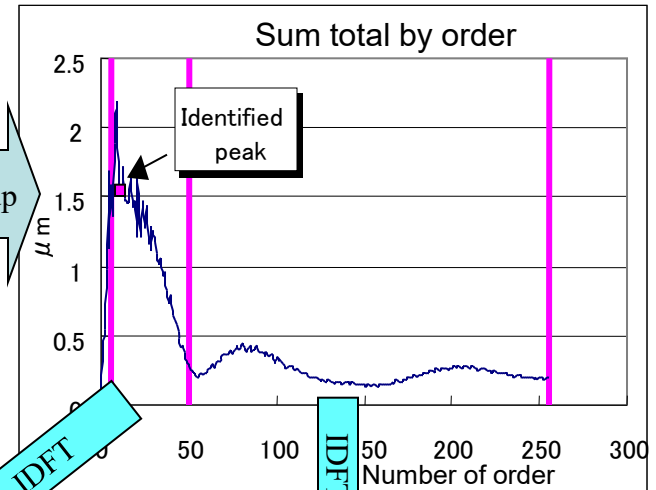
Peening treatment on piston material
Fine ball#400 3.5kgf/cm² 30sec

2D-DFT, we can get a doughnut shape spectrum.

As a result of sum up, we identify a peak close to 11th order.
Fourier synthesize with up to 50th : play back whole of dimple
→ Spectrum up to 50th holds the characteristics of dimple.



Sum up



IDFT

IDFT

1) 2D-DFT with measurement profile
→ Out put of 2D-spectrum

2) Sum up by distance from origin
(By order number)

3) Peak detect: smoothing/differentiation
In this case

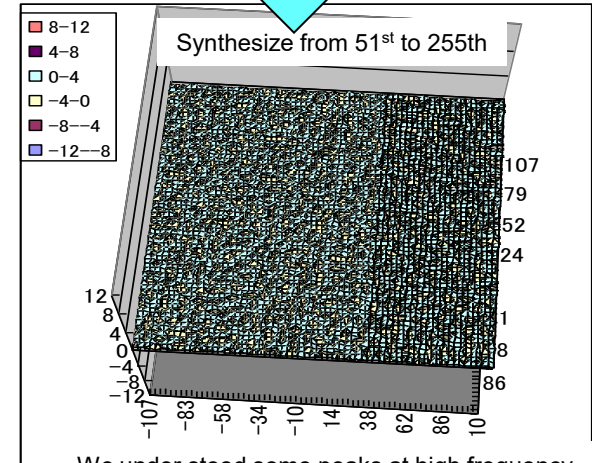
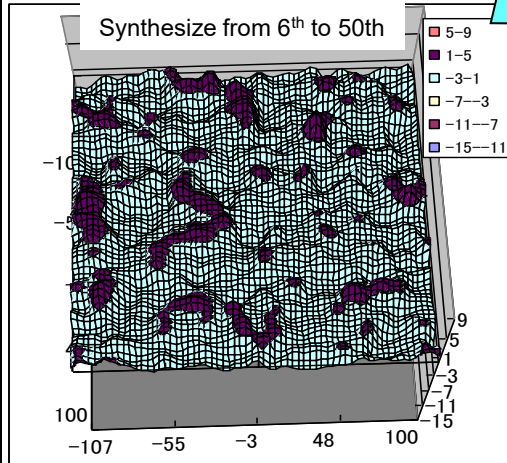
peak order: 11th → Wave length = size/order
= 440/11 = 40 μm

peak height = 1.54 μm

“Index” of dimple

Mean dimple distance : 40 μm

Mean dimple depth: 1.54 × 2 = 3.08 μm

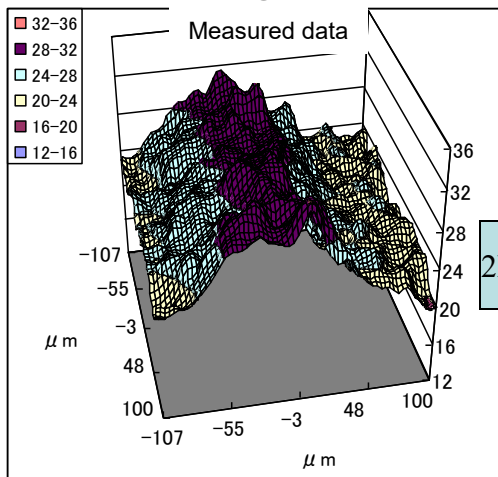


We understood some peaks at high frequency were only noises (not signal).

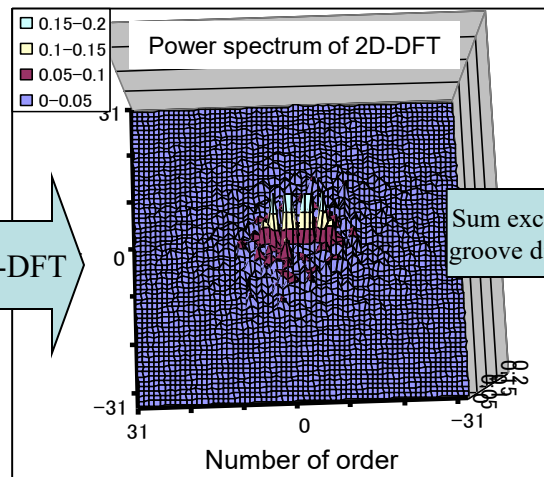
Analysis : Separation of different features

Large grooved surface and small dimples

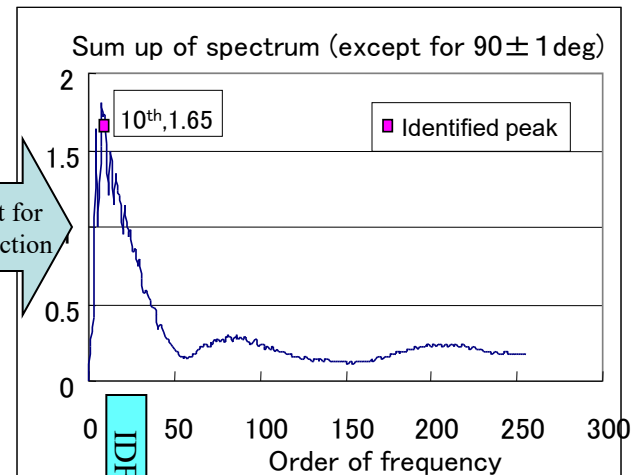
Machining grooved surface + shot peening
Fine ball#400 3.5kgf/cm² 30sec



2D-DFT



Sum except for groove direction



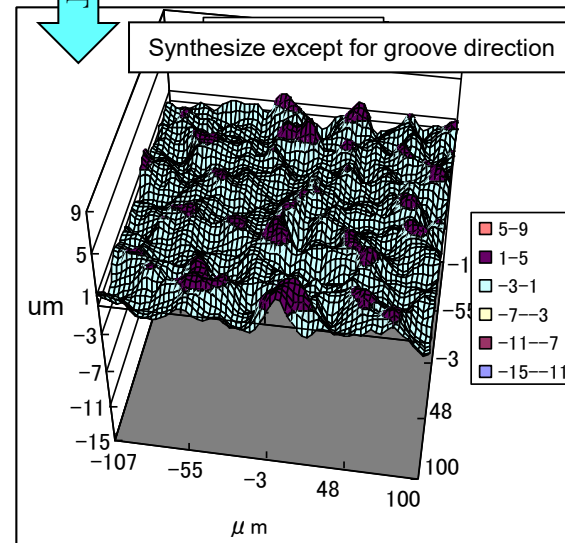
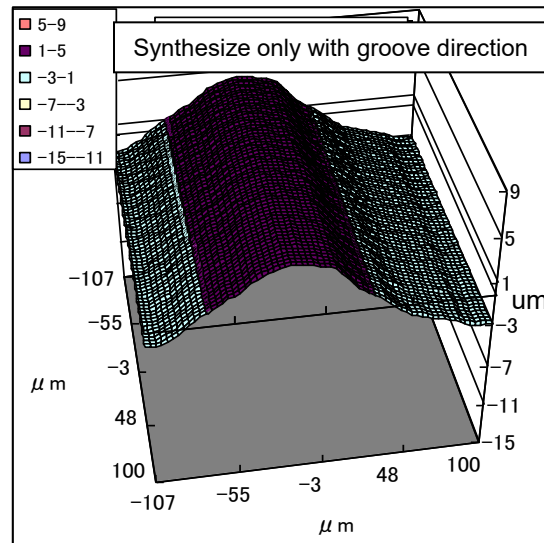
- 1) Out put of DFT
Spectrum with straight and low frequency (Machined groove)
Spectrum with circular shape (dimple)

- 2) Sum up except for groove direction
($90 \pm 1 \text{ deg}$) Eliminate the groove

- 3) Peak detect: smoothing/differentiation method

"Index" of dimple

Mean dimple distance : $440/10 = 44 \mu\text{m}$
Mean dimple depth : $1.65 \times 2 = 3.30 \mu\text{m}$



Investigation on characteristics of dimples

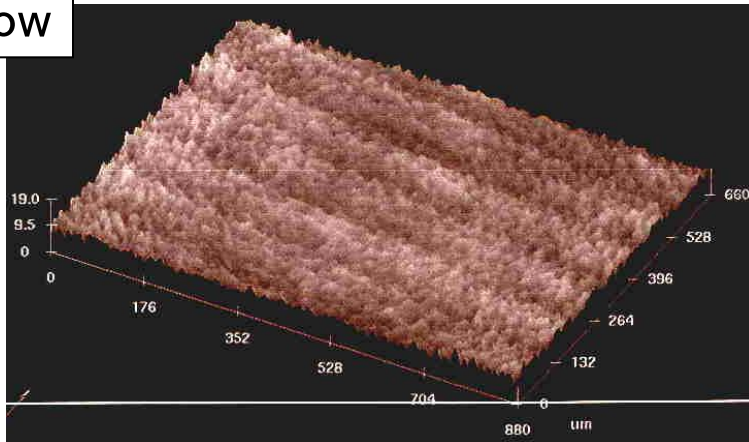
Relationship between shot conditions and dimple index

	Low	Mid low	Mid high	High	Large
Medium	#400	<-	<-	<-	#300
Pressure	1.5 _{kgf/cm2}	2.5 _{kgf/cm2}	3.5 _{kgf/cm2}	4.5 _{kgf/cm2}	4.5 _{kgf/cm2}
Time	12s	<-	<-	<-	30s

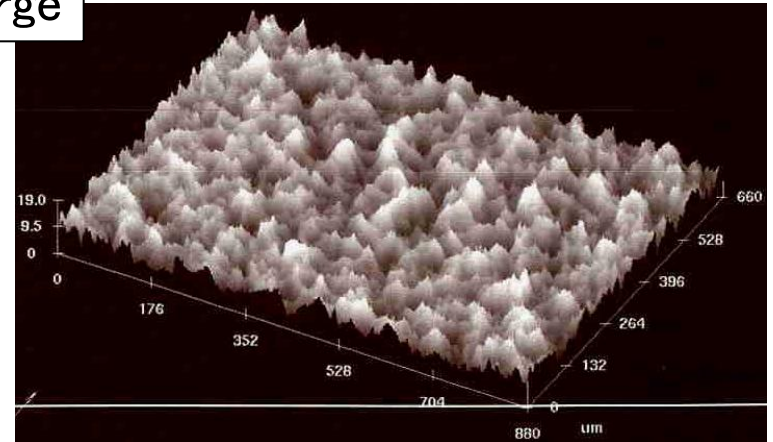
As the intensity of shot condition rises, size of dimple is enlarged. Shot intensity and dimple index (distance and depth) show good relation as the figure.

By using 2D-DFT, we can evaluate surface texture with same indexes, even if base material (Al and steel) or shot medium is different.

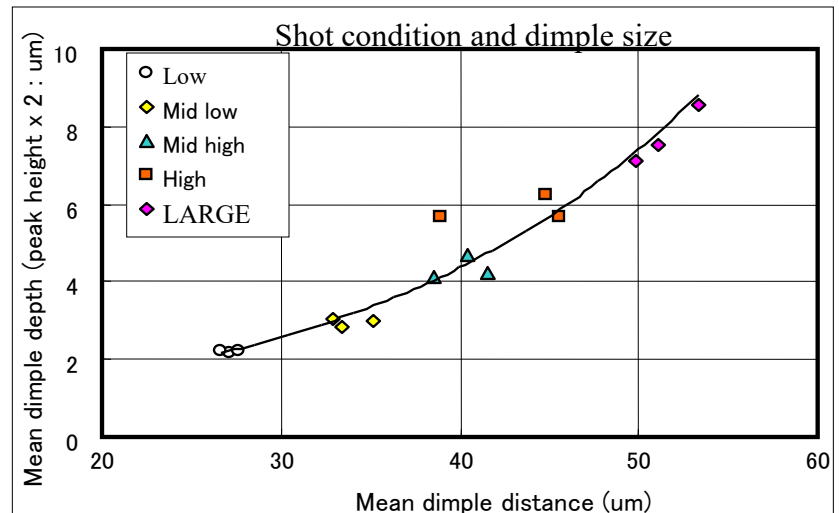
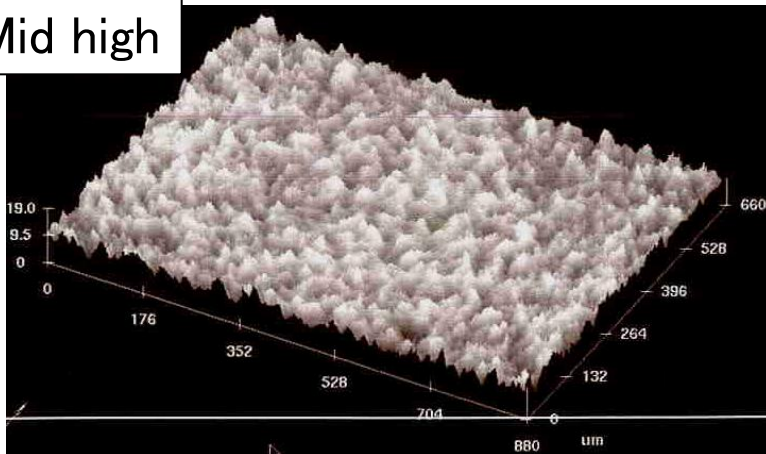
Low



Large



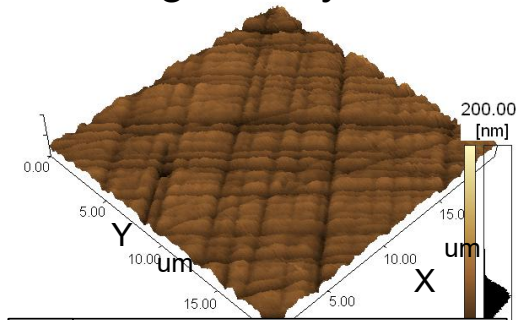
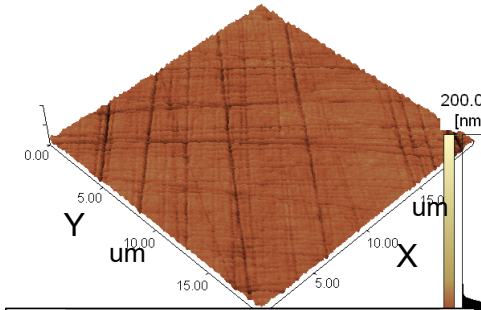
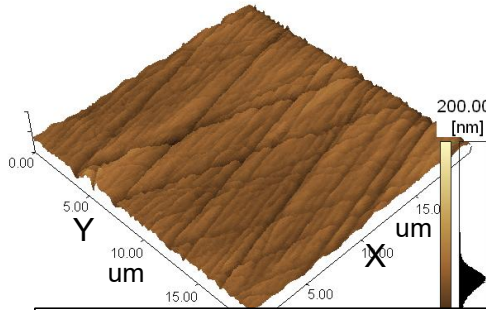
Mid high



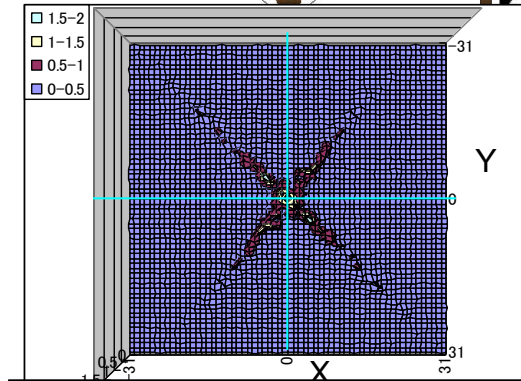
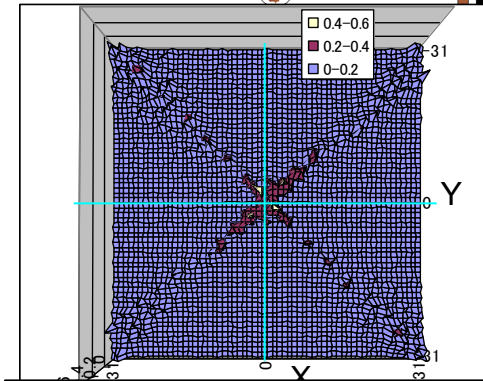
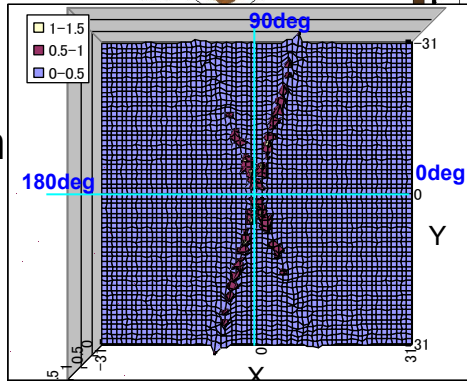
Investigation on cross-hatch with 2D-DFT

Characteristic of cross-hatch (angle, depth, density) can be investigated by DFT

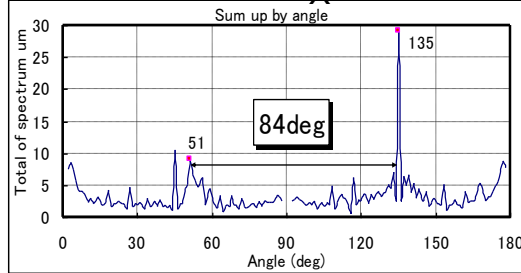
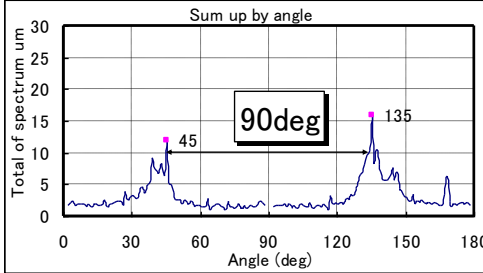
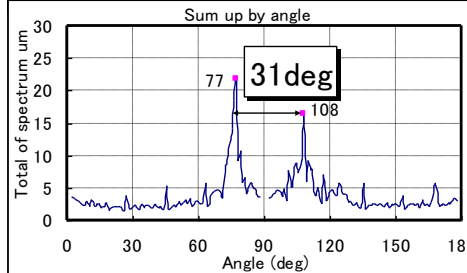
Surface image



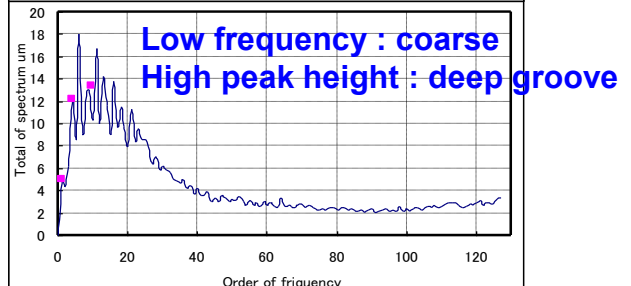
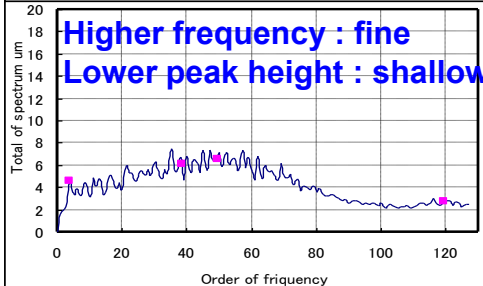
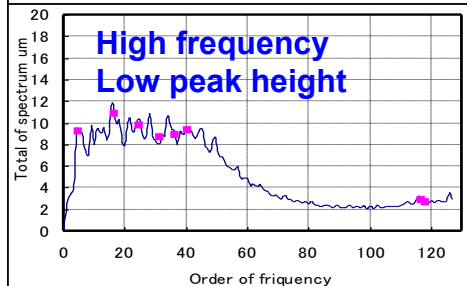
Power spectrum



Sum up by angle



Sum up by frequency



2 dimensional discrete Fourier transform (2D DFT)

Conclusion

- 1) 2D-DFT is useful method to analyze periodicity of all sort of 2 dimensional signals. By using this method, we can investigate on characteristics of surface texture.
- 2) However, there is still some outstanding problem in the reduction of noise (filter). Consistent procedure is necessary to establish some standards with 2D-DFT, like the standard for surface roughness.